

Firmware and Mechanical 101 — polish, box, and what else

Lecture 3 • Fri 2 Oct 2026 • 14:20–16:20 • R311 IAMS

Your block's driver and phone panel, in sim • one alarm rule • your box from the template • what else this can do

Where we are

- The five zones are merged; the board is **on order** since Mon 28 Sep, back $\approx$ 12 Oct, **fully assembled**. Here is the 3D view.
- Today everything runs in **sim mode on your dev board**. The driver you write today runs unchanged on the real board in the wrap-up.
- **The demo checklist** (week of 26 Oct, 5 minutes): ① a waveform out, seen on the scope *and* on the instrument's own input • ② change it from the phone • ③ alarm $\rightarrow$ relay $\rightarrow$ notification • ④ **your block**, its panel, one thing it does, your number.

A block is one class with five methods

```
class OutputsBlock : public Block {  
    const char* name() const override { return "b3"; }  
    void begin() override;                // pins, SPI – once  
    void loop() override;                 // poll; fast; no delay()  
    bool handle(JsonObjectConst cmd, JsonObject reply) override; // one command  
    void status(JsonObject out) override; // your numbers, 20x per second  
};
```

Rules: talks only to **its own hardware** • everything runs from `loop()` — no tasks, no locks, **no** `delay()` • `#ifdef SIM` fakes the hardware • real code is `TODO` until measured

- pins only from `include/pins.h`.

Three data paths on one WebSocket

Command → reply (JSON)

```
{"block":"b3","cmd":"sine","args":  
{"ch":1,"freq":1000,"amp":5}}  
→ {"ok":true,"result":{"..."}}
```

Status (JSON, 20 Hz)

every block's numbers → the strip chart, the alarm engine

Samples (binary, PROTOCOL § 6)

stream {ch, rate_hz, chunk} → rolling frames

capture {ch, rate_hz, n, trig} → one burst frame

→ the **Scope** tab: roll / single / auto, cursor, CSV

Rolling: tens of kS/s (WiFi). Burst: up to the ADC (500 kS/s).

`instrument.py` speaks all three from a PC. Your E2 `press` command was already path one.

One round-trip, end to end

```
button.onclick → api.send('b3','sine',{ch:1,freq:1000,amp:5})  
  → JSON text frame over ws://192.168.4.1/ws  
    → main.cpp queues it → loop() → registry → OutputsBlock::handle()  
      → writes the DAC (or, in SIM, remembers the setpoint)  
    ← reply {"id":7,"ok":true,"result":{"ch":1,"mode":"sine"}}  
  ← promise resolves → toast  
  
every 50 ms: OutputsBlock::status(out) → {"ao1":..., "mode1":"sine"}  
  → broadcast → panel.onStatus(st) → the number on your screen
```

You will explain this to the tutor **before your second flash**.

One panel module per block

```
export default {
  id: 'b3', title: 'Outputs',
  render(el, api) { // once, when the tab opens
    el.innerHTML = `...controls...`;
    el.querySelector('#sine').onclick = () =>
      api.send('b3', 'sine', { ch: 1, freq: 1000, amp: 5 }).catch(e => api.toast(e.message));
    el.querySelector('#alarm').onclick = () =>
      api.addAlarm({ block: 'b3', key: 'ao1', op: 'gt', threshold: 9.5, action: 'notify' });
  },
  onStatus(st) { // 20x per second, your block's status object
    els.ao1.textContent = st.ao1.toFixed(3);
  },
};
```

Every status key you show must exist in your `status()`. Vanilla JS, no build, no CDN — the board is offline.

SIM mode and the alarm engine

SIM (`-DSIM=1`): your `#ifdef SIM` branch returns plausible, *moving* fake values — a slow sine, a counter, a drift towards the setpoint. Write it **first**; flash; see your tab; then the panel; then the real branch.

Alarms are data: `{block, key, op, threshold, action}` tested $20\times$ per second on the same status; fire once on the rising edge; `notify` toasts every phone, `relay:` `<n>:on|off` acts on B4. Persist in `/alarms.json` .

Your one rule — B1 `ai1 > 9 → relay 1 off`

- B2 `v3v3 < 3.0 → notify` • B3 `ao1 > 9.5`
- B4 opto count jump • B5 TRIG edge.

E11 — driver + panel in sim (40 min now, finished at home)

1. Copy `blocks/template/` → `blocks/b<N>_<name>/` ; rename the class; `name()` → `"b<N>"`
2. **SIM branch first** — your status keys, your commands, moving fake values → flash `esp32s3-sim` → your tab shows them
3. `panels/template.js` → `panels/b<N>.js` — one control per command, one readout per key → `uploadfs` → reload
4. **One alarm rule** from a button; trigger it in sim; watch the toast (and B4's relay flip)
5. Explain one round-trip to the tutor
6. **Pair up:** B3 ↔ B1 (a sine into an input) • B5 ↔ B4 (a TTL line into an opto)
7. Branch `b<N>-fw-<name>` → PR; CI builds both envs

Your block page lists your commands and keys: B1 `read_all set_range` • B2 `rails` • B3 `set_dc sine off` • B4 `relay`

Break — 5 minutes

Then: boxes.

Boxes for electronics — six things to know

1. **FDM tolerance:** holes **+0.3 mm**, slots +0.2 mm; a 3.0 mm screw wants a 3.3 mm hole.
2. **Walls 2 mm**; corners rounded; no unsupported overhang $> 45^\circ$.
3. **Heat-set inserts** for M3 — print the boss, melt the insert in.
4. **Layer direction** is strength direction: print the lid flat; stand-offs vertical.
5. **The PCB front panel** makes the cutouts trivial: one rectangle, four holes. The SMAs and the OLED live on the panel PCB, not on the box.
6. **STEP from KiCad** (*File* $\rightarrow$ *Export* $\rightarrow$ *STEP*) — the template already contains the merged board.

Laser-cut acrylic is the alternative for flat panels. Not used for the box.

E12 — your box (30 min now, finished at home)

1. Open the instructor's **Onshape template** → *Copy workspace* into your account
2. The board STEP is in it; the three parameters — **height, wall, foot style** — are in the *Variables* table
3. Change **one thing** that makes it yours: a vent pattern, an embossed name, different feet, a colour note. **One.**
4. *Export* → *STL* (binary, mm) → `docs/students/<name>/box.stl` → commit, with a screenshot

The instructor prints all five on the Bambu the week after. Boards back $\approx$ 12 Oct go straight into them.

What else this can do

- **Bot alerts** — the alarm engine's `notify` → a PC script → LINE / Telegram webhook
- **The Scope tab on a real signal** — and `instrument capture` straight into NumPy
- **Scripted sweeps** — `instrument send b3 sine ...` in a loop, `instrument stream b1.ai1 -
-CSV`
- **OTA** — update the firmware over WiFi, no cable
- **Videos, slides, reports with AI** — the lab's `show-your-work` skills (`education-video` , `slide-deck` , `technical-report` , `showoff-render`): *2-minute live demo*
- **Your own lab's instrument** — the same shape: blocks, one socket, a phone panel, a Python client

Homework 3 — 3 hours • then the wrap-up • then the demo

	h	Due
E11 driver + panel PR — all commands, one value plotted, the alarm rule, CI green; ring review comment-only (V6 first)	1.8	Fri 9 Oct
E12 <code>box.stl</code> in your folder	0.75	Fri 9 Oct
V6 + V7, chapter 3 reading	0.45	

Wrap-up (1.5 h, week of 19 Oct): flash the real board without SIM • your panel moves real hardware • **one measured number** with its method into `SPEC.md` • 60-second video.

Demo (week of 26 Oct): the four-item checklist, 5 minutes, video as the safety net.

Score: block works 40 • PR quality incl. your review 30 • demo clarity 20 • box 10.

Build it. Ship it. Log it.

You did it on a pendulum, on a dev board, on a class board, in firmware, in a box.
The next instrument is yours.